

Name _____

Date _____

10.2 Logic circuits

Getting started

Draw each of the logic gate symbols.

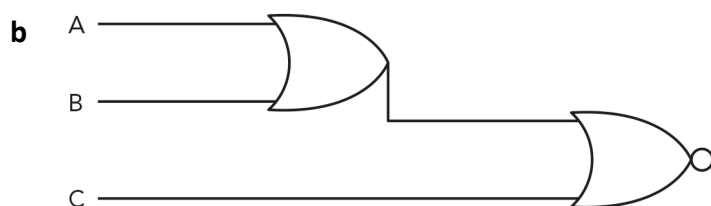
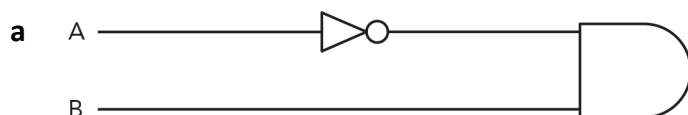
- 1 AND
- 2 OR
- 3 NOT
- 4 NAND
- 5 NOR
- 6 XOR

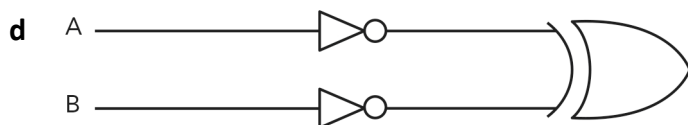
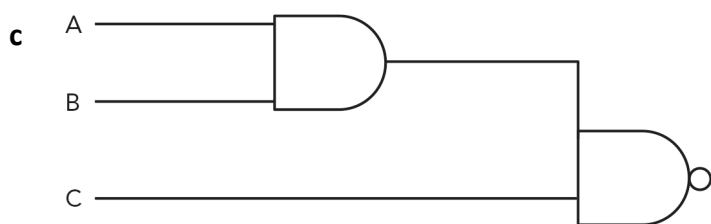
Practice

1 Draw the logic circuit for the following logic statements:

- a A AND B
- b A OR NOT B
- c A AND B AND C
- d (A XOR B) OR C
- e A NAND B NAND C

2 Write the logic expression for the following logic circuits





Challenge

Write the logic statement for each of the truth tables **and** draw its logic circuit.

1

A	B	C	Output
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

2

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

3

A	B	C	Output
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

4

A	B	C	Output
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1